

MILESTONE 1

Client Design Review

Lesika Data Analytics (Potchefstroom)

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Student Number	29290473
Module	CMPG 325
Client ID	CLI-144
Addressing Block	192.168.60.0/24
Technical Challenge	Network Troubleshooting (Fault Isolation)
Change Request	CR15 – Second Internet Connection for Resilience

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1. Client Requirements

1.1 Business Scenario

Lesika Data Analytics is a small data-analysis consultancy based in Potchefstroom. The company takes client data, cleans it, analyses it, builds dashboards, and produces reports. It is a SOHO (Small Office / Home Office) business with 20 employees, expected to grow in 18 months. The organisation requires a secure, scalable, and manageable network that supports wired and wireless connectivity, shared services, and dual Internet connectivity for resilience.

1.2 Employee Headcount

Role	Number
Managing Director	1
Operations / Finance Manager	1
Admin / HR Assistant	1
Data Analysts	8
Data Scientists / Engineers	4
IT / Network Support	2
Client Liaison / Sales	3
Total	20

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1.3 Client Requirements List

1. Support 20 employees now, but scalable
2. Provide wired and wireless network access.
3. Separate staff, management, servers, voice, guest, and IoT/CCTV traffic using VLANs.
4. Provide DHCP and DNS automatically.
5. Provide shared file storage and backup capability.
6. Allow secure Internet access using NAT.
7. Protect company data from guests and unnecessary internal access.
8. Be easy to troubleshoot and document.
9. Use Cisco Packet Tracer for implementation.
10. Be manageable on a SOHO budget.
11. Integrate a second Internet connection for resilience (CR15).
12. Reserve addressing for a branch office that may be opened within 18 months.
13. Demonstrate Network Troubleshooting (fault isolation) as the assigned challenge.

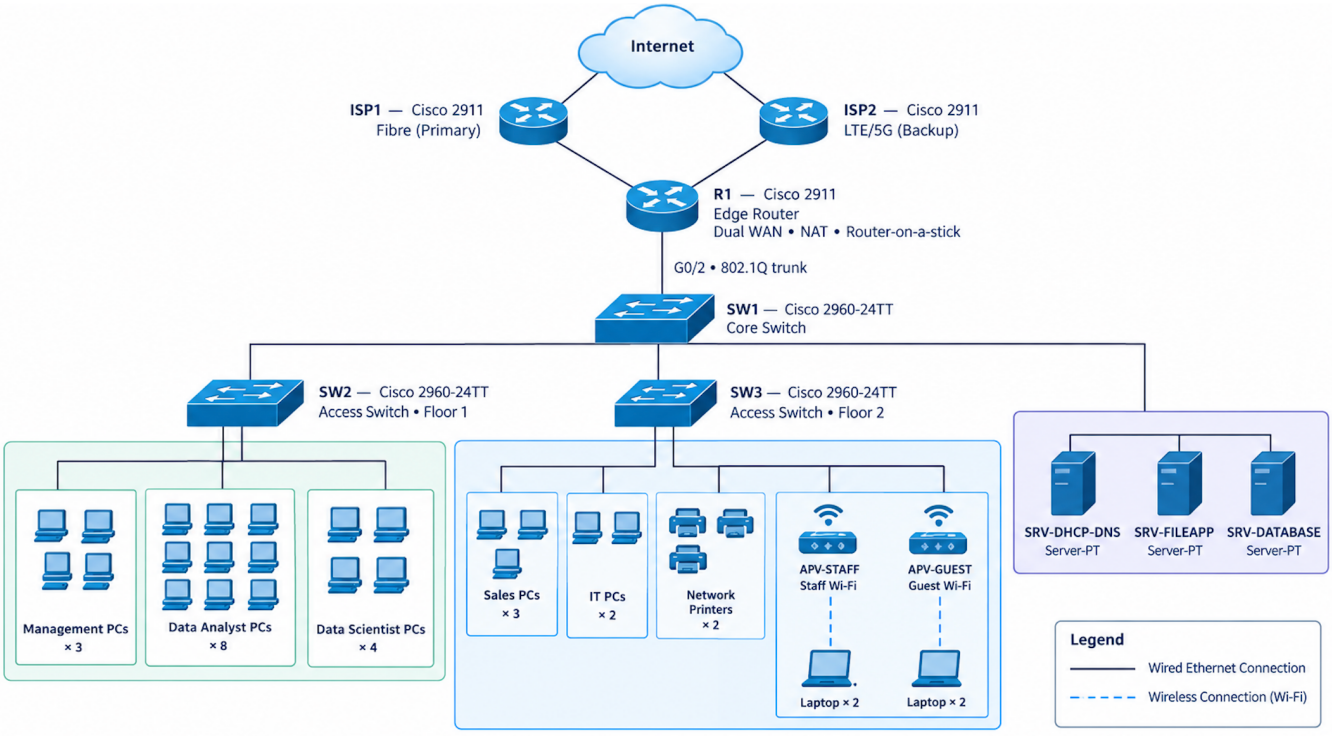
1.4 Devices Required

Device	Quantity	Purpose
Management PCs	3	MD, Ops, Admin
Data Analyst PCs	8	Daily analysis work
Data Scientist PCs	4	Heavier data work
Sales PCs	3	Client liaison

IT PCs	2	Network and server management
Laptops	2	Wireless staff and guest testing
Network printers	2	Shared printing
File / DHCP / DNS server	1	Core services
Database server	1	Structured data
Access Points	2	Staff and Guest Wi-Fi
Edge Router	1	Dual WAN, NAT, inter-VLAN routing
ISP routers	2	Simulated Fibre and LTE/5G providers
Core Switch	1	Trunk to access switches
Access Switches	2	Floor 1 and Floor 2 access

2. Physical Topology

2.1 Physical Layout



2.2 Device Roles

Device	Model	Role
R1	Cisco 2911	Edge router, dual WAN, NAT, router-on-a-stick
ISP1	Cisco 2911	Simulated Fibre ISP
ISP2	Cisco 2911	Simulated LTE/5G ISP
SW1	Cisco 2960-24TT	Core switch (trunk to SW2 and SW3, servers)
SW2	Cisco 2960-24TT	Access switch for Management, Analysts, Scientists
SW3	Cisco 2960-24TT	Access switch for Sales, Printers, IT, APs
SRV-DHCP-DNS	Server-PT	DHCP and DNS services
SRV-FILEAPP	Server-PT	File and application services
SRV-DATABASE	Server-PT	Database server
AP-STAFF	AccessPoint-PT	Staff Wi-Fi

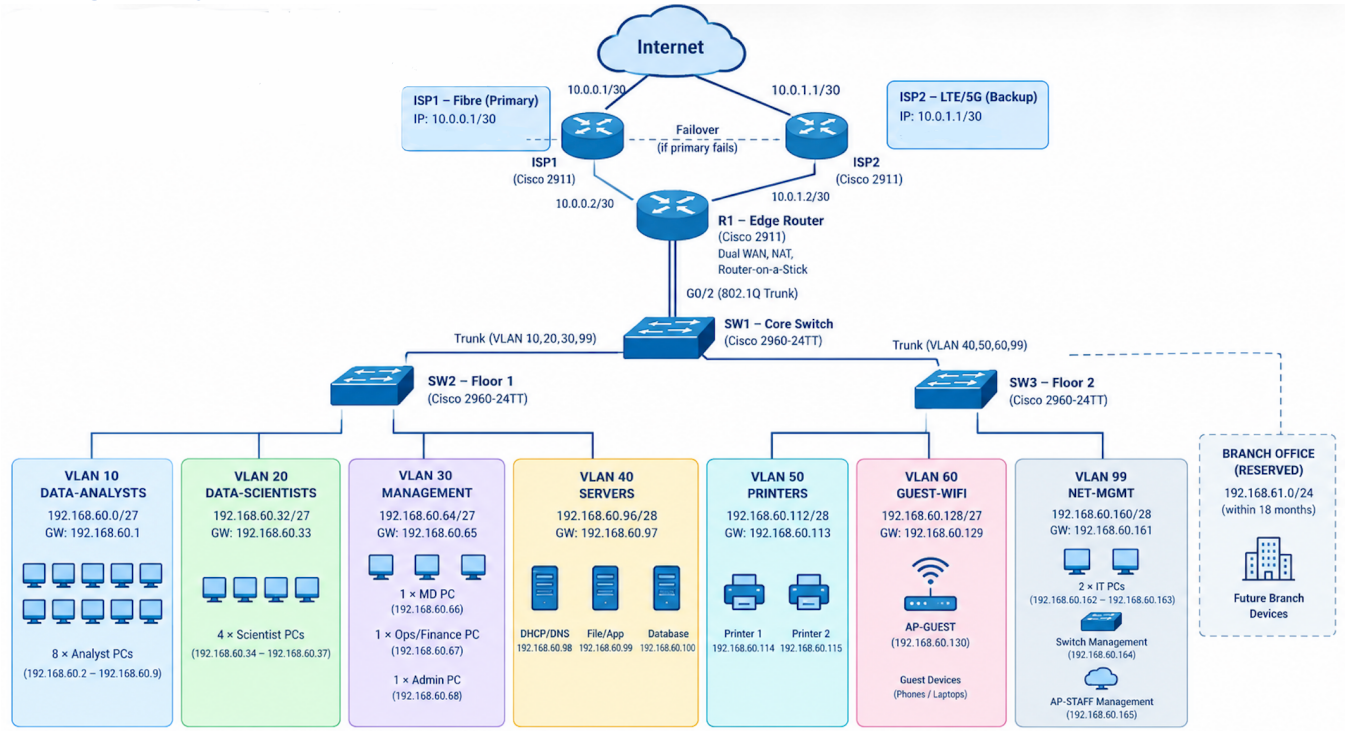
AP-GUEST	AccessPoint-PT	Guest Wi-Fi

3. Logical Topology

3.1 VLAN Plan

VLAN	Name	Purpose	Subnet	Gateway
10	DATA-ANALYSTS	Data analyst PCs	192.168.60.0/27	192.168.60.1
20	DATA-SCIENTISTS	Data scientist PCs	192.168.60.32/27	192.168.60.33
30	MANAGEMENT	Management and Sales	192.168.60.64/27	192.168.60.65
40	SERVERS	File, DHCP, DNS, DB	192.168.60.96/28	192.168.60.97
50	PRINTERS	Network printers	192.168.60.112/28	192.168.60.113
60	GUEST-WIFI	Guest wireless	192.168.60.128/27	192.168.60.129
99	NET-MGMT	Switches, IT PCs, APs	192.168.60.160/28	192.168.60.161

3.2 Logical Layout



3.3 How Devices Communicate

- Devices in the same VLAN communicate through their switch.
- Devices in different VLANs communicate through R1 using router-on-a-stick.
- Internet traffic goes R1 → NAT → ISP1 (primary) or ISP2 (backup).
- DHCP requests are forwarded by R1 to 192.168.60.98 using ip helper-address.
- Guest Wi-Fi is isolated from all internal VLANs by an ACL.

4. IP Addressing Plan

4.1 Assigned Block

192.168.60.0/24 gives 256 addresses, 254 usable, 1 network, 1 broadcast.

4.2 Subnet Calculation

Each subnet is sized using VLSM based on current and future host needs.

VLAN	Hosts needed	Chosen size	Usable
10	25	/27	30
20	25	/27	30
30	25	/27	30
40	8	/28	14
50	4	/28	14
60	25	/27	30
99	8	/28	14

4.3 Full Addressing Table

VLAN	Subnet	Network	Usable	Broadcast	Gateway
10	/27	192.168.60.0	.1 – .30	.31	.1
20	/27	192.168.60.32	.33 – .62	.63	.33
30	/27	192.168.60.64	.65 – .94	.95	.65
40	/28	192.168.60.96	.97 – .110	.111	.97
50	/28	192.168.60.112	.113 – .126	.127	.113
60	/27	192.168.60.128	.129 – .158	.159	.129
99	/28	192.168.60.160	.161 – .174	.175	.161

4.4 WAN Addressing

Link	Network	R1 side	ISP side
R1 – ISP1	10.0.0.0/30	10.0.0.2	10.0.0.1
R1 – ISP2	10.0.1.0/30	10.0.1.2	10.0.1.1
ISP1 public	203.0.113.0/30	—	203.0.113.1
ISP2 public	198.51.100.0/30	—	198.51.100.1

4.5 Static IP Assignments

Device	IP
R1 G0/2.10	192.168.60.1
R1 G0/2.20	192.168.60.33
R1 G0/2.30	192.168.60.65
R1 G0/2.40	192.168.60.97
R1 G0/2.50	192.168.60.113
R1 G0/2.60	192.168.60.129
R1 G0/2.99	192.168.60.161
SRV-DHCP-DNS	192.168.60.98
SRV-FILEAPP	192.168.60.99
SRV-DATABASE	192.168.60.100
PRINTER-01	192.168.60.114
PRINTER-02	192.168.60.115
PC-IT1	192.168.60.162

PC-IT2	192.168.60.163

4.6 DHCP Ranges

VLAN	Range
10	.2 – .30
20	.34 – .62
30	.66 – .94
60	.130 – .158

4.7 Growth

Reserved space after 192.168.60.176:

- 192.168.60.176/28 – Main Office Growth
- 192.168.60.192/26 – Future Branch
- 192.168.60.240/30 – Main–Branch WAN link
- 192.168.60.244/30 – Reserved

4.8 VLAN and Subnet Relationship

One VLAN = one subnet. Devices in the same VLAN communicate directly. Devices in different VLANs must be routed through R1.

5. Initial GitHub Repository

https://github.com/KatDEv98/Final_Year_CMPG325_Networking_Project_Lesika_Data_Analytics_Network

6. Design Justification

Design Decision	Reason
VLAN segmentation	Security, performance, logical separation
Router-on-a-stick	Cost-effective inter-VLAN routing for a SOHO
Dual WAN (CR15)	ISP resilience; floating static route for failover
DHCP with ip helper-address	Centralised IP management across VLANs
DNS on SRV-DHCP-DNS	Name resolution for internal services
NAT overload	Private IPs share a single public IP for Internet
Guest Wi-Fi isolation	Protect internal resources from visitors
Reserved /26 for branch	Meets the 18-month branch office constraint
Fault isolation focus	Assigned challenge – layered troubleshooting

7. Fault Isolation Considerations

The design supports fault isolation in the following ways:

- Each VLAN has a distinct subnet and gateway
- Central router (R1) is the single point for inter-VLAN routing
- Trunks between switches are clearly documented
- Servers have static IPs for reliable testing
- DHCP relay is used per VLAN, isolating DHCP faults
- Network management VLAN 99 isolates infrastructure

Five fault scenarios are planned for Milestone 2:

14. Disabled trunk port between SW1 and SW2
15. Missing VLAN on a trunk
16. Wrong IP address on a PC
17. Missing ip helper-address on a VLAN sub-interface
18. Missing DNS record on the DNS server

End of Milestone 1 Document

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